Cardiac Deformation Lab

Advanced Statistical & Quantitative Analytics Report

Study Title:	**Longitudinal Research Cohort**	**Subject Identifier:**	STAT-P002
Study ID:	3b39f5f5-29f5-4f3a-bb09-f8ed911c8911	**Session Count:**	**5** analyses
Analysis Scope:	**Full Image**	**Detected Outliers:**	0 observations (1.5xIQR)
Report Date:	2026-10-16 05:37 UTC	**Computational Engine:**	NumPy / Farnebäck Dense Flow

1. Descriptive Deformation Statistics (Univariate)

Metric	N	Mean	Median	Std Dev	Min	Max	P25	P75	IQR
Mean Displacement (px)	5	2.8627	3.0283	1.6718	0.7244	4.7954	1.6903	4.0751	2.3848
Max Displacement (px)	5	18.3473	21.3507	12.9964	3.0000	33.4836	6.9999	26.9025	19.9026
Anisotropy Index	5	0.4547	0.4669	0.0652	0.3695	0.5250	0.4078	0.5041	0.0962
Strain Entropy	5	1.5328	1.4243	0.6480	0.8836	2.2881	0.9527	2.1154	1.1627
Strain Gradient	5	0.1573	0.1544	0.1083	0.0317	0.2933	0.0741	0.2331	0.1590
Motion Coherence	5	0.8987	0.8964	0.0051	0.8942	0.9066	0.8955	0.9009	0.0054
Quality Control Score (/100)	5	100.0000	100.0000	0.0000	100.0000	100.0000	100.0000	100.0000	0.0000

2. Longitudinal Change & Standardized Differences

Tracked Metric	First	Latest	Delta (Δ)	Rel Change	Std Diff (d)	Trend
Mean Displacement	0.7244 px	4.0751 px	3.3507 px	+462.5%	1.2350	**Increasing**
Max Displacement	3.0000 px	26.9025 px	23.9025 px	+796.8%	1.1622	**Increasing**
Anisotropy Index	0.3695	0.5041	0.1346	+36.4%	1.2550	**Increasing**
Strain Entropy	0.8836	2.1154	1.2318	+139.4%	1.0187	**Increasing**
Strain Gradient	0.0317	0.2331	0.2013	+634.6%	1.1456	**Increasing**
Motion Coherence	0.8964	0.9009	0.0045	+0.5%	0.5048	**Stable**
Quality Control Score	100.0000 /100	100.0000 /100	0.0000 /100	+0.0%	0.0000	**Stable**

3. Metric Pearson Correlation Analysis

Metric A	Metric B	Pearson r	N Pairs	Association Type
Mean Displacement	**Anisotropy Index**	+0.9980	5	Positive Association
Mean Displacement	**Strain Gradient**	+0.9943	5	Positive Association
Max Displacement	**Anisotropy Index**	+0.9926	5	Positive Association
Mean Displacement	**Max Displacement**	+0.9924	5	Positive Association
Max Displacement	**Strain Gradient**	+0.9890	5	Positive Association
Anisotropy Index	**Strain Gradient**	+0.9868	5	Positive Association

Strain Entropy	**Strain Gradient**	+0.9864	5	Positive Association
Mean Displacement	**Strain Entropy**	+0.9730	5	Positive Association

Note: Correlation describes statistical association only and does not imply causation.

4. Computational Statistical Outliers (1.5×IQR)

No computational outliers identified using the 1.5×IQR boundary rule (or insufficient N < 4 observations).

5. Statistical Warnings & Protocol Consistency

- **Warning:** Longitudinal physical-unit comparison may be unreliable because calibration settings differ between analyses.
- **Warning:** Physical-unit statistical comparison may be unreliable because calibration settings differ.

6. Research Methodology & Interpretation Disclaimer

Computational Analytics Scope: Descriptive parameters, consecutive change differentials, standardized computational differences (effect sizes), and Pearson correlation matrices reflect algorithmic evaluations of image-space deformation fields under classical continuum kinematics. Statistical outliers reflect mathematical dispersion (1.5×IQR) and do NOT indicate patient health or clinical abnormality.

RESEARCH PROTOTYPE ONLY — NO CLINICAL DIAGNOSTIC ASSERTIONS:
These statistics summarize computational image-derived measurements and should not be interpreted as clinical findings. This software does NOT provide medical diagnoses, prognosis, treatment monitoring, or clinical risk assessments. This document is intended strictly for mathematical and biomechanical research experimentation.